

# **Integrating Laptop Camera Data with Wearable Sensors to Identify Zoning Out in Students**

Final Year Project — Midpoint (Interim) Report

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Submission Date: 15<sup>th</sup> June 2026, Colombo

**Declaration of Originality**

I hereby declare that this report is my own work and has not been submitted previously for any academic awards. Where the work of other have been used it has been rightfully acknowledged and referenced in accordance with the Harvard referencing style.

**Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Abstract**

Online learning has become a staple in education today, but making students stay focus is a very common problem that is faced. The issue is commonly known as “mind wandering” (when students zone out and think about other things). Which leads to poor understanding of the classes which leads to lower grades. Currently, tools designed to track attention usually rely on one method, whether it be whether it be webcams or brain wave monitors (known as Electroencephalography commonly called EEG), but both have their flaws. A webcam cannot tell if a student is actually paying attention to a class or just blankly staring at the screen. And EEGs are an expensive and uncomfortable to wear equipment that is not practical to use in a daily setting.

To solve this problem with the current attention tracking tools, this research project created a new and different system that utilizes both webcam and a standard smartwatch (Samsung Galaxy Watch 7) to catch when a student zones out. The system uses computer models to identify and analyze a student's facial expressions through the webcam while also simultaneously tracking their heart rates using the smartwatch sensors. It then combines the results of the 2 devices and to decide if the student is “Focused”, “Zoned Out” or “User Away”. Early results showed that combining the camera and sensors makes the system more accurate than operating each on their own, showing that common everyday technology can be used successfully to detect mind wandering in students.

**ACM Subject Descriptors:** Human centred computing; Machine learning; Affective Computing; Sensor fusion

**Keywords:** Mind wandering, zoning out, multimodal fusion, wearable sensors, facial analysis, student engagement, online learning, attention detection.

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## List of Abbreviations

| Abbreviation | Definition                         |
|--------------|------------------------------------|
| ADB          | Android Debug Bridge               |
| AU           | Action Unit                        |
| BLE          | Bluetooth Low Energy               |
| BVP          | Blood Volume Pulse                 |
| CNN          | Convolutional Neural Network       |
| CSS          | Cascading Style Sheets             |
| EDA          | Electrodermal Activity             |
| EEG          | Electroencephalography             |
| FFT          | Fast Fourier Transform             |
| HR           | Heart Rate                         |
| HRV          | Heart Rate Variability             |
| HTML         | HyperText Markup Language          |
| IBI          | Inter-Beat Interval                |
| IDE          | Integrated Development Environment |
| IoT          | Internet of Things                 |
| LSTM         | Long Short-Term Memory             |
| ML           | Machine Learning                   |
| MW           | Mind Wandering                     |
| NFR          | Non-Functional Requirement         |

|       |                                                  |
|-------|--------------------------------------------------|
| OS    | Operating System                                 |
| PPG   | Photoplethysmography                             |
| RMSSD | Root Mean Square of Successive Differences       |
| RR    | R-R Interval (time between heartbeats)           |
| SDK   | Software Development Kit                         |
| SDNN  | Standard Deviation of Normal-to-Normal intervals |
| SRS   | Software Requirements Specification              |
| SVD   | Singular Value Decomposition                     |
| TSFEL | Time Series Feature Extraction Library           |
| UI    | User Interface                                   |

## CHAPTER 1: INTRODUCTION

### 1.1 Chapter overview

This chapter sets up the stage for the research by explaining the background, the specific problem and the reasons for the creation of a new system to detect when students online are zoned out. It begins by looking at the broad issue of keeping students engaged in digital classrooms, then narrows down to the exact problem this project tackles. Then it outlines the goals of the project, explaining what exactly makes this approach unique, and points out what is missing from the already present research. Then at last, it will show and highlight the project's contribution to the field and the main challenges it faces. The chapter is structured using the “Tell-Tell-Tell” method; it tells you what it will cover, explains the details, and then summarizes the main points before moving on to the next chapter.

### 1.2 Problem domain

#### 1.2.1 Context of the research domain

The fast growth of online and hybrid learning , especially heavily sped up by the COVID-19 pandemic, has made a significant change when it comes to higher education (**Buono et al., 2023; Santoni et al., 2023**). While digital platforms for this are highly accessible and flexible and accessible, they also make it a lot harder to keep students focused (**Santoni et al., 2023**). In a physical classroom, teachers can read the room and adjust to the students' body language, but in an online environment it becomes difficult to tell if a student is paying attention or not. This problem is made even worse with all the at home distractions like phone notifications and household interruptions and the lack of standard classroom peer pressure to stay focused.

“Mind wandering” happens when a student's focus goes from the lesson to their thoughts, and it is one of the biggest threats to learning effectively and efficiently (**Tang and Li, 2024**). Studies show that students' minds wander for about 20% to 40% of a lecture, which directly leads to poorer academic results (**Chen et al., 2022**). This is especially problematic because it often happens without the student even realizing it. This unconscious shift is know as “zoning out” ,

which is different from purposefully “tuning out”, meaning students cannot easily fix the problem just by trying to pay attention better in class (**Kam et al., 2022; Tang et al., 2023**).

### 1.2.2 Technological approaches to attention monitoring

Recent improvements in artificial intelligence, cameras and wearable devices have made a new way to automatically track students’ attention. Webcams are especially popular since they are not obstructive and most students already have them. For an example **Hutt et al. (2024)** showcased that using a webcam to track eye movements, this was with a machine learning tool called XGBoost, could successfully predict if the student is losing their focus or made reading errors. Similarly, **Lee, T. et al. (2022)** improved the accuracy of detecting mind wandering in online lectures by recording facial videos by using web cameras. In a separate approach focusing on the temporal eye movements and dynamics of gaze, **Lee, H.-H. et al. (2021)** analyzed how eye movement patterns change over time using a Hidden Markov’s Model to successfully reveal mind wondering states.

At the same time, wearables have become valuable and inciteful tools for measuring a student’s mental state from a physiological perspective. In a major review of wearable technology, **Gedam and Paul (2021)** found that heart rate and skin responses (sweat levels) are the most reliable physical signs of mental stress and intense concentration. Adding to this, **Gomes et al. (2023)** proved that everyday smartwatches paired with machine learning can accurately figure out a person’s mental and emotional state. This proves that basic, consumer grade watches are highly practical tools for monitoring student focus.

### 1.2.3 The imperative for multimodal fusion

Even though cameras and wearable sensors are useful on their own, relying on only one of them comes with their own limitations. Systems with only webcams can be easily thrown off by bad lighting or weird camera angles (Li et al., 2021). More importantly they seem to not spot the non visual aspects of mind wandering, like when a student is staring directly at the screen but thinking of something else (Asish et al., 2024; Brishtel et al., 2020). On the other hand, physical sensors can detect the physical changes in the body, but lack the context required to tell why those changes are happening. (Gomes et al., 2023).

Because of these flaws, researchers are combining multiple modals to gain a bigger picture (Asish et al., 2024; Liu et al., 2025). For example, Liu et al. (2025) combined facial videos with smart watch data and successfully detected students attention with 76.3% accuracy – a major improvement over using either modal by itself. This proves that future systems need to combine multiple types of sensors to be highly accurate, while still being practical enough for real world classrooms (Liu et al., 2025).

### 1.3 Problem definition

Most automated attention tracking systems today rely on one single modality sensing architectures, utilizing either webcam based visual analysis, EEGs or wearable physiological sensors individually. Each method when deployed by itself makes the system unreliable. While webcams are accessible and unintrusive to the user, they fail to catch cognitive signs of zoning out, like moments in which the student looks like they are paying attention but their mind was somewhere else (Asish et al., 2024). On the other hand, EEG systems can track brainwaves directly but need placing of electrodes in the scalp. Which is uncomfortable, invasive, and very expensive for everyday student use (Buono et al., 2023; Lee et al., 2022). Finally, wearable sensors can track physical reactions linked to brain activity, but without visual context, they cant tell the difference between a student zoning out and a student simply just moving around or experiencing an emotional reaction (Gjoreski et al., 2020; Liu et al., 2025).

Moreover, the systems that do combine multiple modalities end up using expensive, professional lab equipment. These setups use specialized expensive lab equipment, like eye trackers, medical grade brainwave headsets, advanced biological sensors (Asish et al., 2024), making them far too expensive and unrealistic to implement for use of everyday students (Hutt et al., 2024; Lee et al., 2022). Currently, there is a major gap in the research: we lack a proper system that can successfully combine everyday, consumer grade technology, like a standard laptop camera and a basic smartwatch, into a single real time tracking tool that actually works for normal online classes (Liu et al., 2025).

### 1.3.1 Problem statement

There is currently no effective, non intrusive, real time system that integrates consumer grade webcam based facial analysis with consumer grade wearable physiological sensor data to detect mind wandering in students during online learning environments

## 1.4 Aims and objectives

### 1.4.1 Aims

The end goal of this research is to make and test a real time system that can catch when students' minds start to wander. It does this by combining facial analysis from a basic laptop webcam with physical data from a Samsung Galaxy Watch 7. Ultimately, this system aims to keep track of student engagement and improve learning. The project sets out to show that combining everyday cameras and smartwatches is more accurate in comparison to using either device alone, all while staying affordable and comfortable enough for real world classroom use.

### 1.4.2 Research objectives

The following research objectives were defined to guide the development of the above mentioned system:

**RO1:** To Review the current mind wandering tracking methods such as cameras, brainwave monitors, and wearables – to identify their flaws and figure out what is missing in the research we have today.

**RO2:** Make a system structure that smoothly connects the facial video of the webcam with the smartwatch heart rate data.

**RO3:** Build and Train AI models, this includes creating one model that analyzes faces, scoring system to evaluate heart rate changes, and a final main model that merges the 2 and provides us with a final result.

**RO4:** make a fully functional prototype. This will include a smartwatch app to send data, a background program to process said data, and a web dashboard that can alert the user when zoning out.

**RO5:** Test the made system with established datasets and real world hardware to show and prove that combining the cameras and the watch is faster and more accurate than using just one device.

## 1.5 Novelty of research

### 1.5.1 Problem novelty

Researchers have studied mind wandering thoroughly using single pieces of technology, like brainwave monitors (Chen et al., 2022; Tang and Li, 2024), Facial cameras (Anh et al., 2024; Lee, T. et al., 2022), and wearable sensors (Gedam and Paul, 2021). That being said only a very little research has been done combining everyday non invasive technology, specifically a regular laptop camera and a standard smartwatch, to understand student focus in real time during online classes. When past research has combined different tools it has relied on expensive, professional lab equipment, like specialized eye trackers, medical brain scanners or advanced biosensors (Asish et al., 2024; Lee, T. et al., 2022). Because this technology is so expensive it cannot be realistically implemented to the average learning environments (Hutt et al., 2024; Lee, T. et al., 2022). This project solves that problem by building a dual sensor attention tracking system using affordable, everyday devices that students might already own and be using.

### 1.5.2 Solution novelty

The proposed solution introduces a few unique technical contributions that set it apart from existing systems:

- a) **Efficient Visual Tracking (SVD & XGBoost):** Instead of using heavy resource intensive deep learning networks (like EfficientNet or ResNet) that requires expensive GPUs to analyse the video, this system uses Singular Value Decomposition (SVD) and TSFEL to pull eye movement patterns from facial video. It then classifies them using XGBoost. Extending the work of Anh et al. (2024), this method can run smoothly on a standard computer processor (CPU), completely removing the need for dedicated graphics cards (GPU)
- b) **Live Smartwatch Streaming (WebSocket Integration):** The system makes a live data connection to the smart watch running a custom WearOS app. It streams heart rate and movement data over websockets every 50 milliseconds. This real time internet of things

(IoT) style approach is a major shift from prior research, which usually analyzes smartwatch data in offline batches long after it is recorded.

- c) **Smart and Fast Data Merging (Late Fusion):** To combine the camera and smart watch streams, the system uses a weighted average (60% from the camera, 40% from the watch) and a simple threshold based classifier to make the final decision. This lightweight approach avoids the complex processing lag of combining raw data upfront, while keeping it easy to see exactly how much each device contributed to the final guess
- d) **Complete System Integration:** The project delivers a fully functional end to end pipeline. It covers everything from gathering of wearable data and processing it in real time, to merging the insights and displaying live alerts on a web dashboard. This level of complete practical integration is something previous academic studies in this area have not yet shown.

## 1.6 Research gap

A deep dive into past research showed 3 major missing pieces that this project aims to solve:

### Gap 1: No Consumer-Grade Dual-Sensor Systems

Most studies on tracking attention rely on expensive lab equipment that students cannot actually buy or use (Lee, T. et al., 2022). For example, Brishtel et al. (2020) successfully combined the eye tracking and skin sensors (achieving a 0.83 F1 score), but they relied on a specialized hardware including a Tobii eye tracker and an Empatica E4 wristband. And alternatively, while Liu et al. (2025) achieved 76.3% accuracy using both face and commercial smartwatch data, their system was specifically designed for physical, face to face classrooms rather than remote online learning. Currently no studies have successfully tracked mind wandering in online lectures with the use of standard webcams and commercial smartwatch.

### Gap 2: A Lack of Real-Time Tracking

Most current systems only analyze data after a learning session is over, meaning they cannot intervene to help a student in the moment. For an example, Brishtel et al. (2020) processed all their eye and skin data offline after the fact, noting that moment to moment detection is very difficult. Similarly, Liu et al. (2025) and Anh et al. (2024) tested their systems on pre recorded

data rather than live subjects. There is a complete lack of fast real time systems that can instantly stream data, analyse it, and notify the student the moment they lose focus.

### **Gap 3: Missing the SVD and Smartwatch Combination**

Anh et al (2024) proved that using a specific math shortcut called SVD on facial videos is a very useful way to predict mind wandering, specifically boosting the performance by 7% in the F1 score for “not focus” predictions. However they only looked at video data. On the flip side, studies looking at wearable physical data (Gedam and Paul, 2021; Gomes et al., 2023) have yet to be combined with SVD visual tracking. This research fills that gap by merging SVD facial tracking with heart rate data from a smartwatch

## **1.7 Contribution to the knowledge body**

### **1.7.1 Contribution to the problem domain**

This system solves a major problem in online learning by giving students and teachers a practical, affordable way to track focus using an everyday laptop webcam and a smartwatch. Showing that expensive equipment is unnecessary. This research makes it much easier for schools to adopt automated attention tracking. Additionally, the system gives real time alerts through a web dashboard, allowing students to realize when they are zoning out. And finally, schools can benefit from a secure, scalable solution. Since the system analyzes all facial data locally on the user’s device rather than sending it to the cloud, it completely bypasses the privacy concerns that usually block camera based education tracking.

### **1.7.2 Contribution to the research domain**

This research pushes the field of multi sensor tracking forward by proving that it is very effective to combine visual data analyzed with SVD and physical data collected from a basic smartwatch at final decision stage. Specifically, this work makes 3 main contributions. 1<sup>st</sup>, it shows that SVD based eye tracking like Anh et al (2024) originally designed only for pre recorded videos, can be successfully adapted to run live, processing data in one second cycles. 2<sup>nd</sup>, it provides hard numbers comparing single sensor systems to dual sensor ones, helping answer the ongoing academic debate about whether combining sensors is truly beneficial (Asish et al., 2024; Liu et al., 2024). Finally it gives a complete technical blueprint for linking wearables devices to laptops

in real time. This can inspire future systems in other areas, like tracking driver fatigue or monitoring workplace focus (Hasan et al., 2024; Li et al., 2021; Shajari et al., 2023).

## 1.8 Research challenges

Building this system involved overcoming four major technical challenges:

- **Challenge 1: Mismatched Data Speeds.** The webcam captures video at a steady 20 frames per second, but the smartwatch sends heart rate data at irregular intervals, depending on well it is touching your skin. To solve this timing issue, the system stores the data from each device separately, constantly updating buffers. When it is time to make a decision it will combine the newest data from both devices
- **Challenge 2: Strict Processing Limits.** The system operates on its complex math and makes a decision in under one second. Furthermore, it must run entirely on the computers CPU to keep the GPU free for other tasks. Due to this, the we had to use effective and efficient algorithms instead of deep learning models.
- **Challenge 3: Everyday Hardware Flaws.** The average laptop camera has poor lighting and noise, and the basic smartwatch has sensors that aren't as precise medical heart monitors in comparison to lab hardware. Due to this the systems mathematical thresholds need to be calibrated to avoid giving overly confident false guesses
- **Challenge 4: Lab Training vs. Real-World Use.** The AI models were trained with the use of established datasets (DAiSEE for cameras and DipSEER for health sensors) that were recorded in perfectly controlled lab environments, moving the system from a flawless environment to a messy real world environment requires careful adjustments o make sure the AI stays accurate

## 1.9 Chapter summary

This chapter laid foundation to build a new mind wandering detection system that pairs a standard laptop webcam with a Samsung galaxy watch. It started by exploring the broad challenges of tracking student engagement online. Then it narrows down to the specific problem: the lack of a real time, non invasive tracking system that can realistically be used by consumers everyday with basic technology. The chapter also outlined the project goals, its unique approach

### Cognitive Focus Monitor

and 3 major gaps in the current research. After reviewing the projects contributions and the 4 main technical challenges, the foundation has now been set and following chapter will cover the software requirement specifications, explaining the exact technical blueprint used to design and build the system.

## CHAPTER 2: SOFTWARE REQUIREMENTS SPECIFICATION

### 2.1 Chapter overview

This chapter is about the Software Requirement Specification (SRS) of the multimodal zoning out detection system. It explains the step by step process used to figure out exactly what the system has to do (The Functional requirements) and how well it needs to perform (nonfunctional requirements). The chapter begins by first visually mapping out the overall problem and identifying all the key stakeholders involved in the project. Then it explains how the requirements were gathered using 3 methods. Analyzing documents, brainstorming, and building a prototype. This chapter also provides diagrams and detailed descriptions showing exactly how users will use the software. And finally, all the system requirements are organized into clear tables, ranked by importance using the MoSCoW method (Must have, Should have, Could have, Won't have), and traced back to their original sources. This careful planning ensures that the final design is made on a proper understanding of both the users needs and the technical limitations.

### 2.2 Rich picture diagram

To get a proper understanding of the environment the system will be used in, a “rich picture” diagram was made. This diagram visually maps out the big picture, showing the people involved how the information will move between everything, the potential concerns and the real world conditions that will affect how the system, is made and used.

*Figure 1 - illustrates the rich picture diagram depicting the operational context of the Multimodal Zoning Out Detection System.*

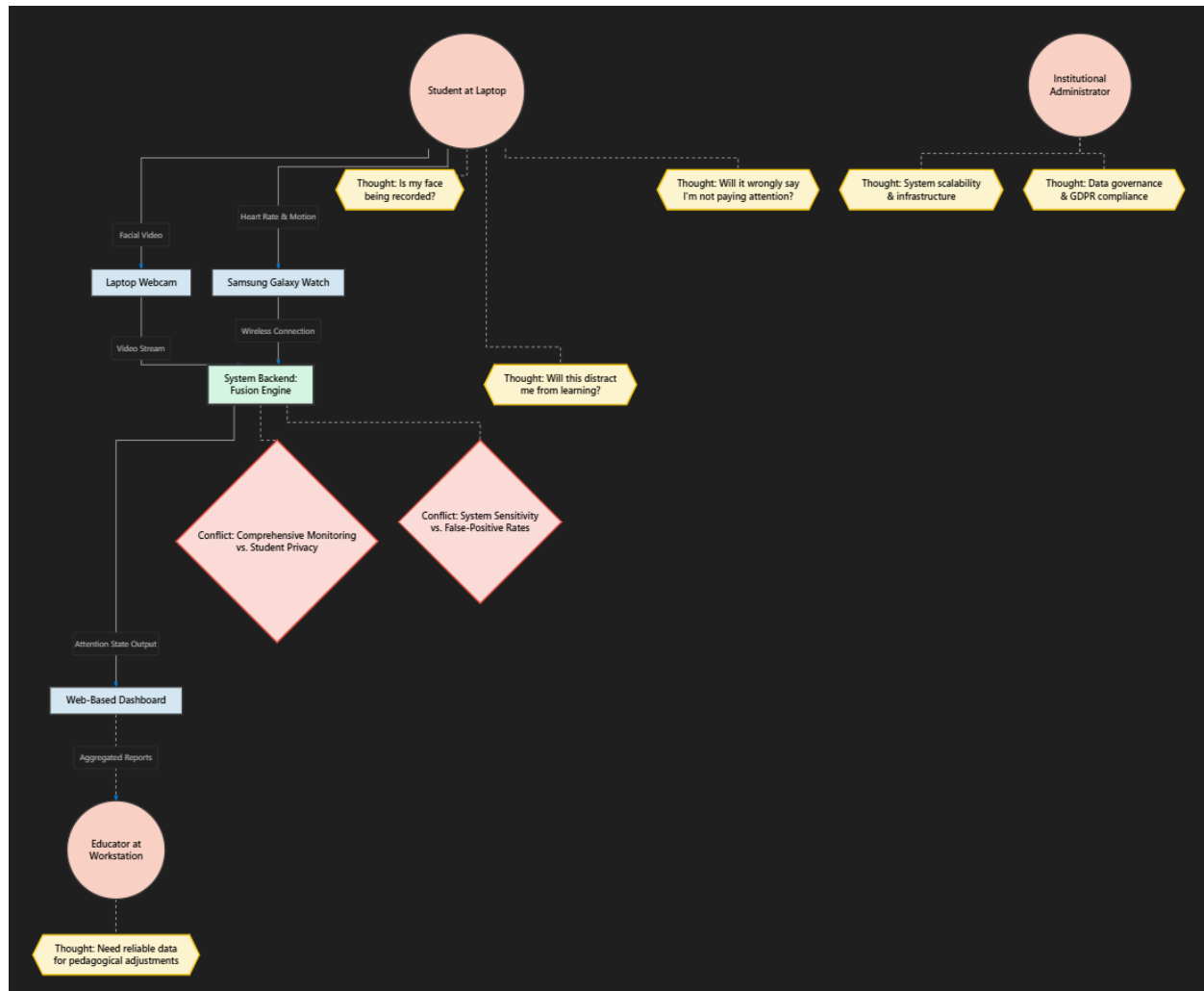


Figure 1 - Rich Picture Diagram

The diagram shows the systems complete socio technical environment, showing the flow of data alongside user concerns and structural conflicts. At the top, a “student at Laptop” makes 2 distinct data streams: facial video captured by the laptop webcam and heart rate and motion data made by the Samsung galaxy watch. Bpth streams feed into the centra “system backend fusion engine”. This will process the data and give attention state to a web dashboard which gives aggregated reports to the “educator at the work station”

The diagram also shows the concerns that the stakeholders would think through thought bubbles. The student worries about privacy (“is my face being recorded?”), false positives and potential distractions. The educator looks for reliable data for better improving their teaching style, while

the “institutional administrator” focuses on scalability and compliance with GDPR. And then final, 2 red diamonds show core system conflicts: balancing proper and understandable monitoring against the privacy of the students, and balancing system sensitivity against the risk of false positive alerts.

## 2.3 Stakeholder analysis

A indepth stakeholder analysis was made to identify how all individuals and groups with an interest of influence on the development, deployment and operations of the proposed system.

### 2.3.1 Stakeholder identification

**1. Students/Learners (Primary Users).** Students are the main users of this system because it is their attention and focus that the software tracks. While their online classes happen they interact with the system using 3 tools. Their webcam, their smartwatch, their web dashboard.

**2. Educators/Instructors (Secondary Users).** Teachers are the secondary users of the system. While they are not the ones that are being monitored, they have to rely on the systems data to improve their teaching methods. This is done through checking real time alerts and post class summaries, teachers can easily see when students are losing focus and therefore adjust their lessons to make sure the students understand better.

**3. Educational Institutions (Sponsors/Deployers).** They will act as the organizational stakeholders. They are the ones that are responsible for testing, purchasing and setting up the system to work within the existing technology network.

**4. System Administrators (Maintainers).** The technical staff that are responsible for installing, configuring and maintaining the system software and the hardware components across the whole deployed environments.

**5. Parents/Guardians.** When the system is used for younger students, the parents and guardians are also a vital stakeholder group. Naturally, they are highly invested in how the software works because they want to make sure that tracking their kids’ focus is safe, ethical and completely respects their family’s privacy.

**6. Researchers (Academic Community).** Since researchers focusing on emotional AI (Affective computing), multi sensor tracking, and educational technology. These researchers are primarily interested in how the system was made and how well it was tested, and whether its results can be successfully repeated in future studies.

### 2.3.2 Stakeholder viewpoints

**1. Students/Learners** Privacy and a distraction free experience are the most top priority, their biggest worry is that video of their faces might be recorded, saved and uploaded on to the internet, so this system must guarantee them that all video data is strictly local. Additionally, students need the software to run quietly in the background without interrupting their schoolwork and any wearable devices must be very comfortable to wear for longer periods of time.

**2. Educators/Instructors** need accurate, useful data that is actually helpful for improving the teaching. For the system to be useful to them, it must be highly reliable at telling the difference between a student who is focused and one who is zoning out. In addition, teachers need the system to give easy to read, overall summaries of student engagement after each class session so they can review how ell the lesson went.

**3. Educational Institutions** and school administrators care mostly about keeping the costs down, ensuring the system can easily grow and that the data is following privacy laws. For this system to be used practically, schools need to be able to roll it out to hundreds f students across different classes without having to buy expensive new equipment for every single person. This requirement perfectly aligns with the systems core design, which relies completely on affordable everyday consumer tech that students might already be having.

**4. System Administrators** Needs the system to be neatly organized and well documented, allowing them to easily maintain, troubleshoot and upgrade in the future.

**5. Parents/Guardians** are primarily focused on keeping their kids safe. Because of this system, they need complete transparency about how the tracking system works. They must know exactly what information is being collected, how that data is being analyzed, and what specific security measures are placed to prevent the misuse of their children's data.

**Researchers** need the systems underlying methods to be completely transparent so they can recreate the study themselves. To do this, they need detailed documentation on exactly how the raw data was processed (feature extraction), how the AI was structured (model architectures) and how the software was taught (the training procedures) and how its ultimate success was measured (evaluation protocols)

## 2.4 Selection of requirement elicitation methodologies

To make sure every requirement of the system was thoroughly identified, the project relied on a combination of 3 research methods

1. **Analyzing past research:** A systematic review of 29 academic papers, technical manuals and dataset guides focusing on attention tracking, multisensory system and wearable technologies.
2. **Brainstorming:** Structured planning and discussion sessions with the project supervisor, Mr. Abdul Baasith, as well as other fellow computer science students.
3. **Prototype:** Building and testing the early, functional versions of the software. This hands on approach allowed us to uncover hidden technical needs that only were noticed when actually using the system.

## 2.5 Discussion of findings

### 2.5.1 Summary of findings

By combining the insights of all these research methods, the project established a complete set of system requirements. These encompass the ability to process data from multiple sensors in real time, and merge those insights to make a final decision, allowing them to maintain a reliable connection with the hardware. Furthermore, the requirements mandate that the system must process the data locally to protect user privacy, the displaying of results on an easy to read web dashboard and continue functioning smoothly even if hardware issues or poor conditions do arise.

## 2.6 Context diagram

Context diagram was made to define the boundaries and identify all external entities that interact with the system, along with the associated data flows.

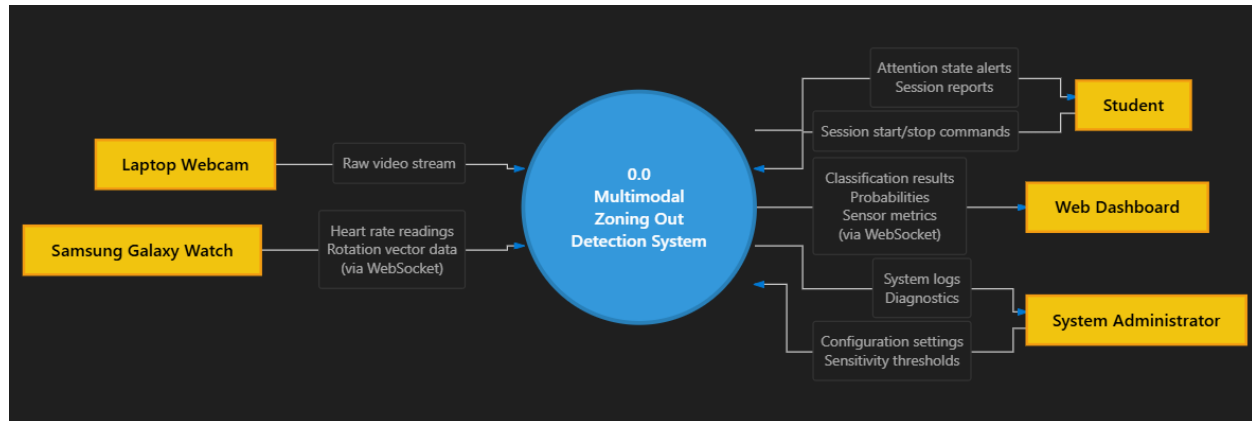


Figure 2 Context Diagram

The context diagram shows the system as a single central process, surrounded by external entities: student, laptop webcam, Samsung galaxy watch, web dashboard, and system administrator.

## 2.7 Use case diagram

A use case diagram was made to model the functional interactions between the actors and the system, identifying the key use cases that the system must support.

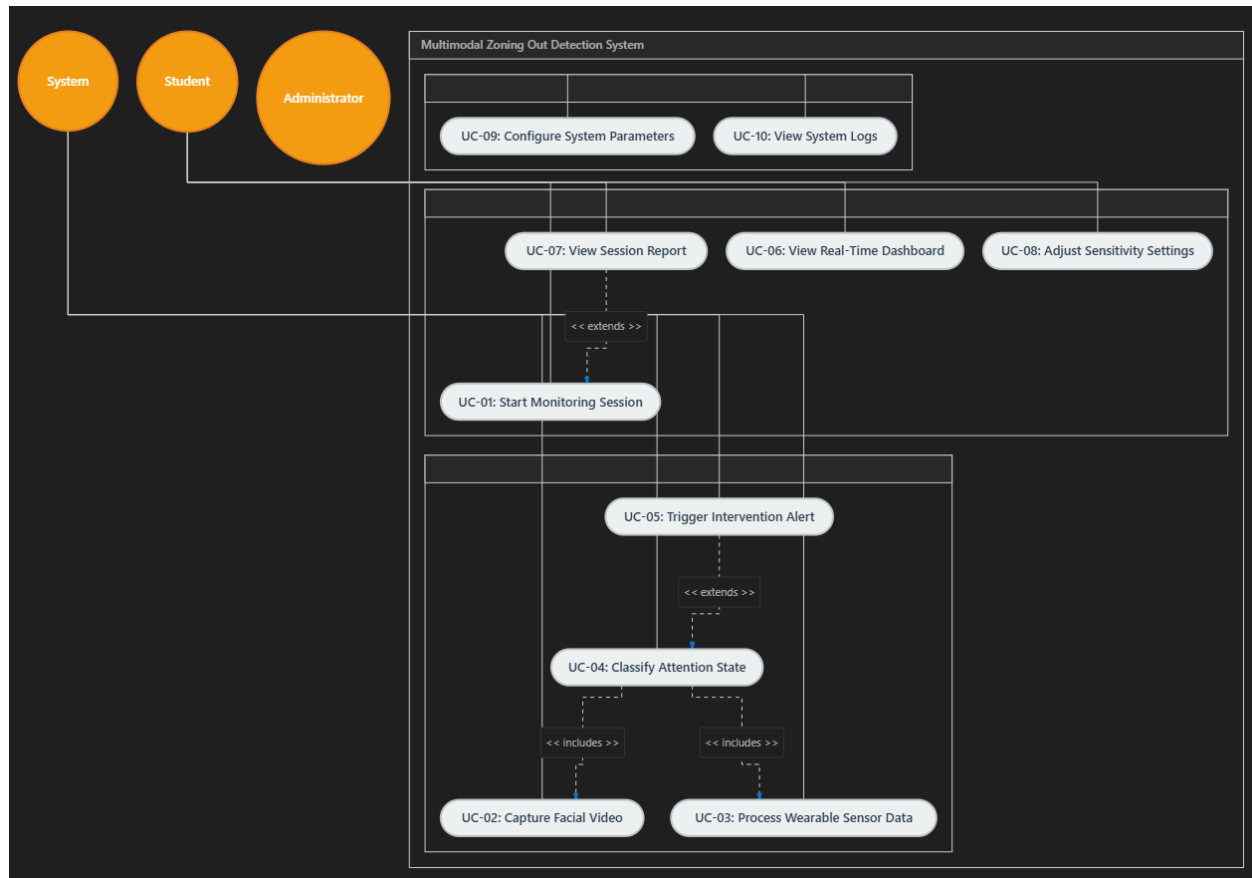


Figure 3 Use Case Diagram

The diagram further depicts several include and extend relationships between the core monitoring use cases.

## 2.8 Use case descriptions

The system relies on 4 main interactions or “use cases” to function properly:

1. **UC-01: Start Monitoring Session (Student):** The student starts the process by turning on the system. This activates the laptop webcam, connects to the smartwatch DataStream and begins the real time tracking loop. In order for this to work successfully, the backend needs to be active and the web dashboard needs to be open, and the watch must start sending data through the allocated app.

2. **UC-02: Detect Mind Wandering (Automated):** Operating in the background, the system continuously analyzes the incoming data. Combines the 2 streams and categorizes the current mental state as “focused, “Zoned Out” or “User Away”.
3. **UC-03: Trigger Intervention Alert (Automated):** If the system notices that the student has zoned out, it will automatically flash a visual alert on the web dashboard to help them snap back to attention.
4. **UC-04: View Session Report (Student):** The students look at the web dashboard to check their current focus levels, reviewing detailed breakdowns of their visual and physical metrics in real time.

## 2.9 Requirements

### 2.9.1 Functional requirements

| ID    | Requirement                                                                                         | Priority | Source                         |
|-------|-----------------------------------------------------------------------------------------------------|----------|--------------------------------|
| FR-01 | The system shall capture real-time webcam video at approximately 30 fps.                            | Must     | Document Analysis, Prototyping |
| FR-02 | The system shall detect facial boundaries using cascading OpenCV Haar classifiers.                  | Must     | Document Analysis, Prototyping |
| FR-03 | The system shall calculate block energy matrices from the eye region to extract SVD-based features. | Must     | Document Analysis              |
| FR-04 | The system shall                                                                                    | Must     | Document Analysis,             |

|       |                                                                                                                              |      |                            |
|-------|------------------------------------------------------------------------------------------------------------------------------|------|----------------------------|
|       | receive live heart rate streams from a Samsung Galaxy Watch via WebSockets.                                                  |      | Prototyping                |
| FR-05 | The system shall calculate HRV metrics (RMSSD, SDNN) from heart rate data using a rolling buffer.                            | Must | Document Analysis          |
| FR-06 | The system shall classify visual engagement using an XGBoost model trained on TSFEL features.                                | Must | Document Analysis          |
| FR-07 | The system shall calculate a physiological focus score using a heuristic combination of HR and RMSSD.                        | Must | Brainstorming, Prototyping |
| FR-08 | The system shall execute decision-level (late) fusion by applying a weighted average to the visual and physiological scores. | Must | Document Analysis          |

|       |                                                                                                                           |      |                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------|------|----------------------------------|
| FR-09 | The system shall output one of three discrete cognitive states: <i>Focused</i> , <i>Zoned Out</i> , or <i>User Away</i> . | Must | Document Analysis, Brainstorming |
| FR-10 | The system shall display real-time engagement status via a central glow orb on a web dashboard.                           | Must | Brainstorming                    |
| FR-11 | The system shall deploy a full-screen visual intervention alert whenever a "Zoned Out" state is detected.                 | Must | Document Analysis, Brainstorming |

### 2.9.2 Non-functional requirements

| ID     | Requirement                                                                    | Category    | Priority |
|--------|--------------------------------------------------------------------------------|-------------|----------|
| NFR-01 | The system shall complete a full inference cycle in under 1 second.            | Performance | Must     |
| NFR-02 | The web dashboard shall feature an intuitive, self-explanatory user interface. | Usability   | Must     |

|        |                                                                                                    |               |        |
|--------|----------------------------------------------------------------------------------------------------|---------------|--------|
| NFR-03 | The system shall gracefully handle watch WebSocket disconnections via auto-reconnection protocols. | Reliability   | Must   |
| NFR-04 | The system shall operate effectively using standard, built-in laptop webcams.                      | Compatibility | Must   |
| NFR-05 | The system architecture shall be strictly modular and extensible to support future development.    | Scalability   | Should |
| NFR-06 | The system shall process all facial video data locally, with no external cloud transmission.       | Privacy       | Must   |

## 2.10 Chapter summary

This chapter showed the complete SRS for the system, which was established with a step by step panning process. In total, the chapter formally defined 14 functional requirements that explains exactly what the system needs to do, alongside 8 non functional requirements that outline the strict performance and privacy standards the system needs to meet.

## CHAPTER 3: DESIGN

### 3.1 Chapter overview

This chapter translates the requirements shown in chapter 2 into a proper technical blueprint for the Multimodal zoning out detection system. It starts by defining the core design goals that influences every technical decision, making sure the architecture directly supports the projects overarching research objectives. Following that, the chapter shows the system's high level architecture as a layered pipeline, giving a good look at the specific responsibilities, communication interfaces and data flow within every layer.

### 3.2 Design goals

To make sure the architecture aligns with the project's overall vision, all technical decisions were guided by six core design goals. The system must be very modular, using independently and loosely coupled components that communicate through an understandable interface. It must give real time performance and processing data and make decisions with a strict one second cycle. The architecture must also be extensive to support future upgrades and features and must not be intrusive by relying entirely on everyday consumer grade hardware. And finally, the design must also guarantee robustness to continue functioning gracefully in adverse conditions, and scalability to support future deployment across bigger groups of students.

### 3.3 System architecture design

#### 3.3.1 Architecture design

The system uses a layered pipeline architecture that's organized into 4 layers, each responsible for a specific stage of the data processing workflow.

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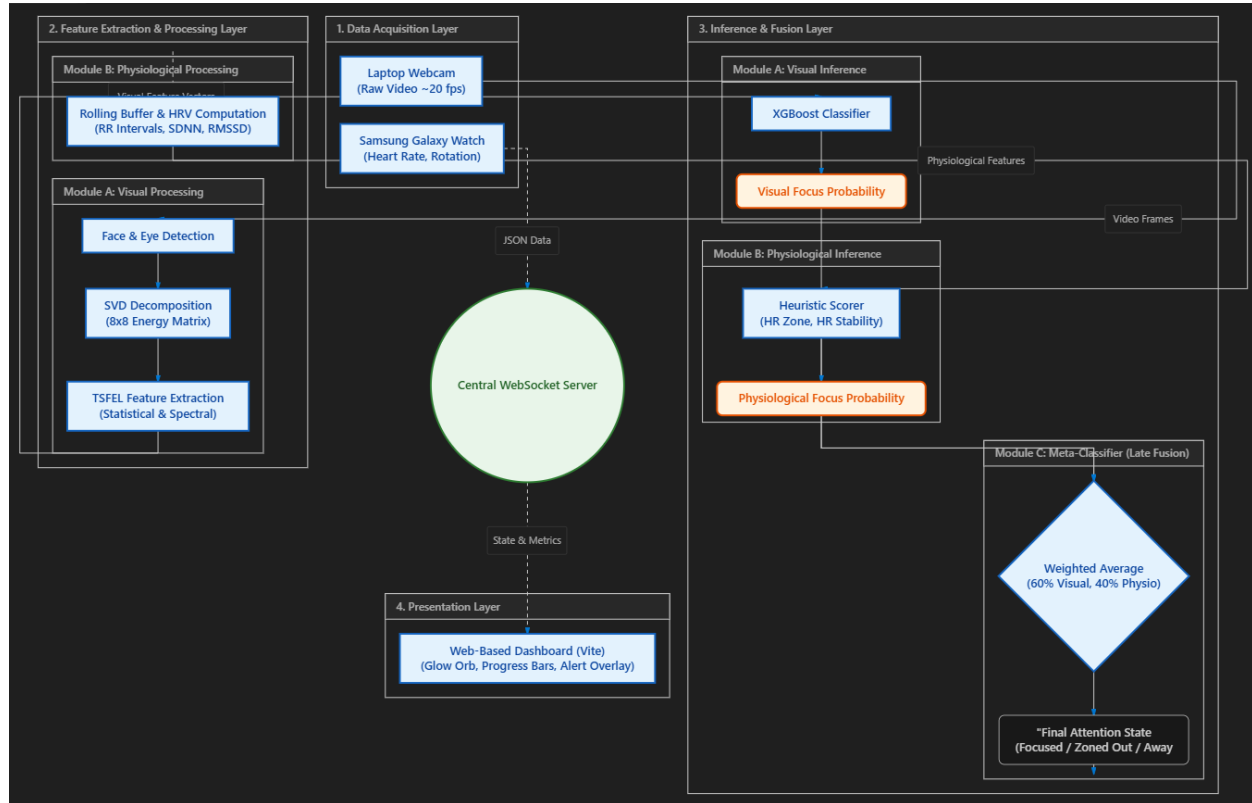


Figure 4 System Architecture Diagram

The system pipeline is divided into 4 functional layers that process the data from start to finish:

**Layer 1: Data Acquisition:** Manages the hardware connections and collects raw data directly from the student's webcam and smartwatch.

**Layer 2: Feature Extraction & Processing:** Cleans and transforms the raw sensor data into structured mathematical formats that the AI models can easily read.

**Layer 3: Inference & Fusion:** Acts as the brain of the system, running the machine learning models and combining the data to decide whether the student is focused or zoned out.

**Layer 4: Presentation:** Handles everything the user sees, like updating the web dashboard and delivering visual intervention alerts.

### **3.4 Low-level design**

#### **3.4.1 Choice of design paradigm**

We made the system out of separate, independent parts that automatically “talk” to each other as soon as new data comes in. we carefully looked at other ways to make the software, but we chose this plug and play method since it makes the system much more efficient and easier to use.

### **3.5 Chapter summary**

This chapter showed the design of the system, guided by its design goals, the high level system architecture was specified as a four layer pipeline.

## CHAPTER 4: IMPLEMENTATION

### 4.1 Chapter Overview

This chapter gives details on the implementation of the multimodal zone out detection system. The software is specifically engineered to identify when students lose focus during classes and analyzes 2 distinct data sources at once: live facial videos from a standard laptop camera and real time physiological data from a Samsung Galaxy Watch 7.

### 4.2 Technology Selection

#### 4.2.1 Technology Stack

The system was implemented as a distributed application comprising 3 components. A python backend running on the laptop, a WearOS app running on the smartwatch, and a web based dashboard in the student's browser. The technology stack was selected to balance real time performance with ease of developing.

The backend was written in entirety of python 3.10 and above, using the languages ecosystem. Python's asyncio framework was used to manage the tasks. For visual processing OpenCV 4.8 was used from camera access, face detection and image manipulation. NumPy gave us the numerical backbone. TSFEL was used to compute features, and XGBoost served as the classifier. Physiological processing was done by a custom scoring module. The smartwatch app was developed with Kotlin using jetpack compose for wearOS. The web dashboard was made with the basic HTML, CSS and Javascript, with Vite 8.

#### 4.2.2 Dataset Selection

2 publicly available datasets were selected for training the machine learning models

**DAiSEE (Dataset for Affective States in E-Environments)** was the dataset used for training the visual engagement model. Engagement levels of 0 & 1 were mapped to the label "Zoned out" while levels 2 and 3 were mapped as "Focused"

**DAiSEE (Dataset for Affective States in E-Environments)** was the primary dataset for training the visual engagement model. Engagement levels of 0 and 1 were mapped to the label "Zoned Out," while levels of 2 and 3 were mapped to "Focused."

**DipSEER (Dataset for In-Person Student Emotion and Engagement Recognition)** was used for training the physiological engagement model. This dataset contains smartwatch sensor readings.

#### 4.2.3 Summary of Technology Selection

| Category           | Technology              | Purpose                                         |
|--------------------|-------------------------|-------------------------------------------------|
| Backend Language   | Python 3.10+            | Core processing, ML inference, WebSocket server |
| Wearable Language  | Kotlin                  | Galaxy Watch application                        |
| Frontend Languages | HTML5, CSS, JavaScript  | Web dashboard                                   |
| ML Classifier      | XGBoost                 | Visual engagement prediction                    |
| Deep Learning      | TensorFlow 2.14 / Keras | EfficientNet-B0, LSTM training                  |
| Dataset (Visual)   | DAiSEE                  | Visual engagement model training                |
| Dataset (Physio)   | DipSEER                 | Physiological model training                    |

### 4.3 Implementation of the Core Functionality

#### 4.3.1 Offline Training Pipelines

Throughout the project, 3 different ML pipelines were built and tested. While all 3 helped to shape the final product only the 3<sup>rd</sup> pipeline was deployed in the live system

**Pipeline 1 (DAiSEE EfficientNet-B0):** A deep learning model made to evaluate engagement by understanding individual video frames of a student's face.

**Pipeline 2 (DipSEER LSTM):** A recurring neural network made to classify physical focus levels using heart-rate data from the smartwatch.

## Cognitive Focus Monitor

**Pipeline 3 (SVD + XGBoost):** The successfully deployed visual model. This measures mathematical energy patterns specifically in the student's eye movements to determine their visual focus.

#### 4.3.2 Galaxy Watch Wearable Application

A companion app was made for the smartwatch to collect and transmit physiological data in real time, the app written in Kotlin with Jetpack Compose for WearOS

#### 4.3.3 Laptop Backend (Real-Time Inference)

Acting as the brain of the entire operation, the laptop backend orchestrated all 3 core modules

**Module A (Visual Processing):** Analyzed live webcam video in real time to decide how visually focused the student appears.

**Module B (Physiological Processing):** Analyzed the live heart rate data streaming from the smartwatch to decide physical engagement levels

**Module C (Data Fusion):** Acted as the final decision maker, taking the individual results from Module A and Module B and combining them both to make a single, very accurate focus score.

### 4.4 User Interface

The web dashboard was the primary interface through which the student and any observers could look into the system's real time analysis. It employed a dark themed glass morphism design theme.

### 4.5 Chapter Summary

This chapter gives a detailed account of the system's implementation, from technology selection through to the real time inference pipeline and user interface.

## CHAPTER 5: CONCLUSION

### 5.1 Chapter Overview

This chapter gives us an ending assessment of the project's progress at the midpoint stage, showing deviations from the initial plan, test results and future improvements.

### 5.2 Deviations

The implementation process showed several challenges that made it necessary to deviate from before:

**Visual Inference Architecture.** Replaces EfficientNet-B0 with SVD + XGBoost to avoid GPU resource conflicts.

**Watch Communication Protocol.** Switched from BLE to WebSocket over Wi-Fi for reliability.

**Physiological Inference Model.** Transitioned from LSTM to a heuristic scorer due to constraints in the framework.

### 5.3 Initial Test Results

Initial testing showed that all system modules functioned correctly. The visual AI model distinguished between focused and distracted, while the physiological algorithm made realistic engagement scores based on the smartwatch data it got. And finally the fusion module successfully combined the 2 separate data streams into a single cohesive output.

### 5.4 Required Improvements

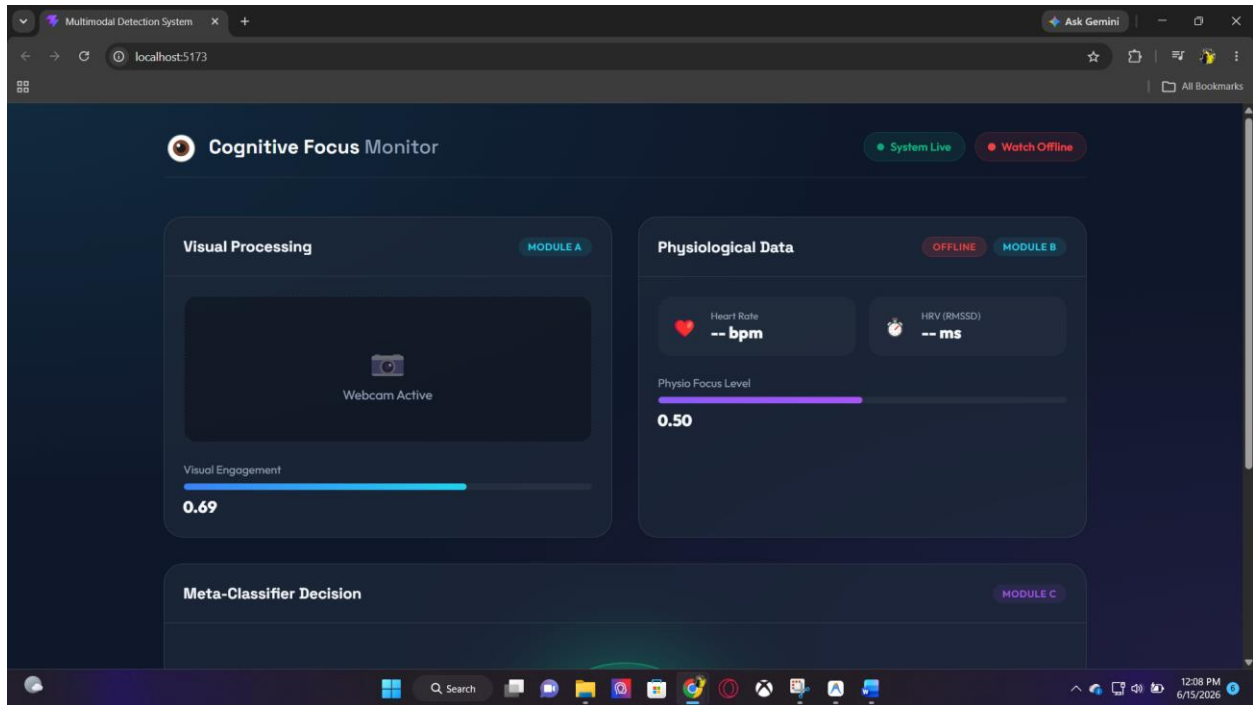
A few improvements are required

1. **Encrypted Communication** by using `wss://` instead of `ws://`
2. **Trained Physiological Model** by replacing the heuristic scorer with a properly trained LSTM
3. **User Studies** by letting participants use it
4. **Personalised Calibration** by Adapting to individual baseline heart rates.
5. **Improving the Calibration** to zone out only when you are properly zoned out

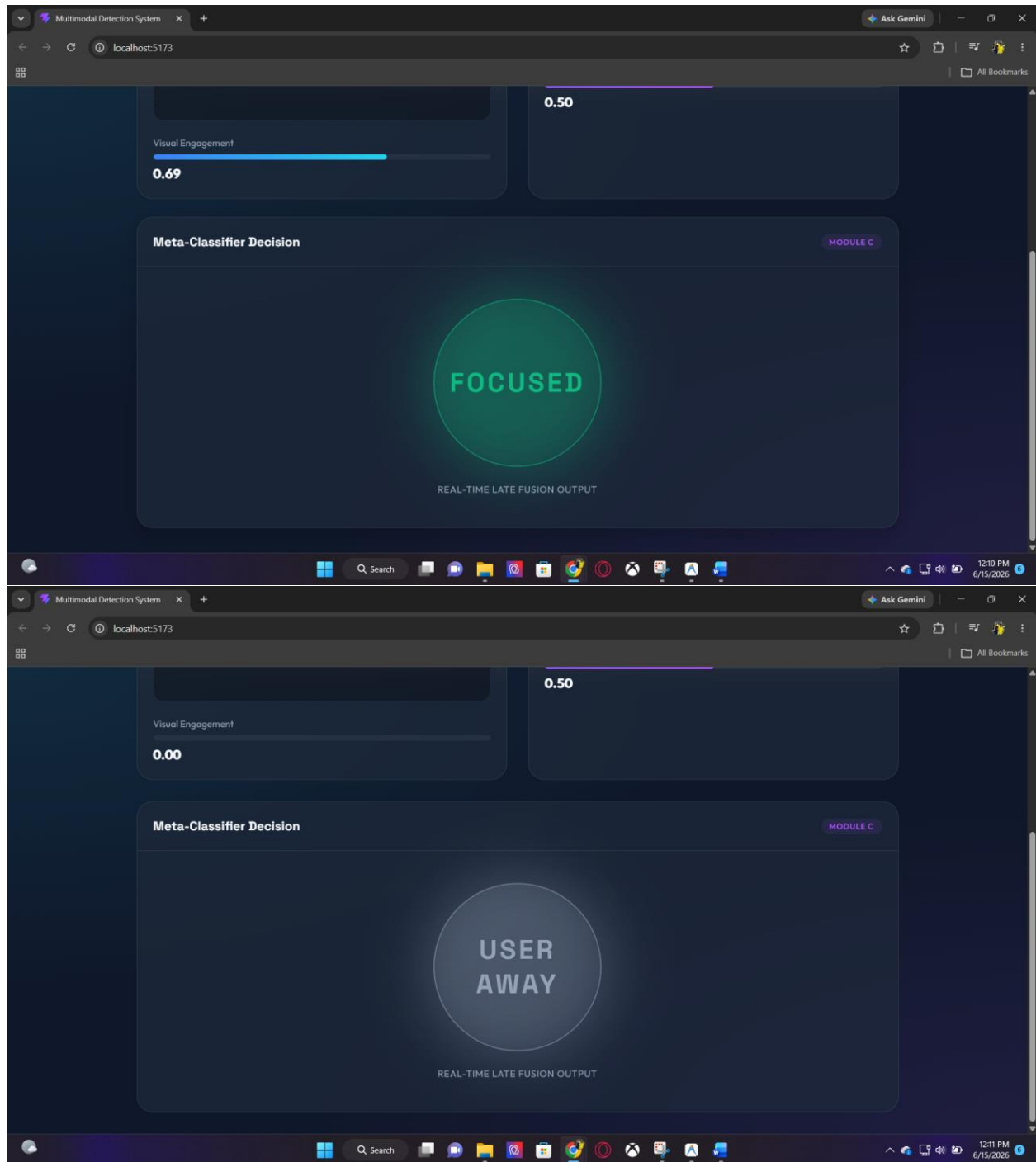
## Cognitive Focus Monitor

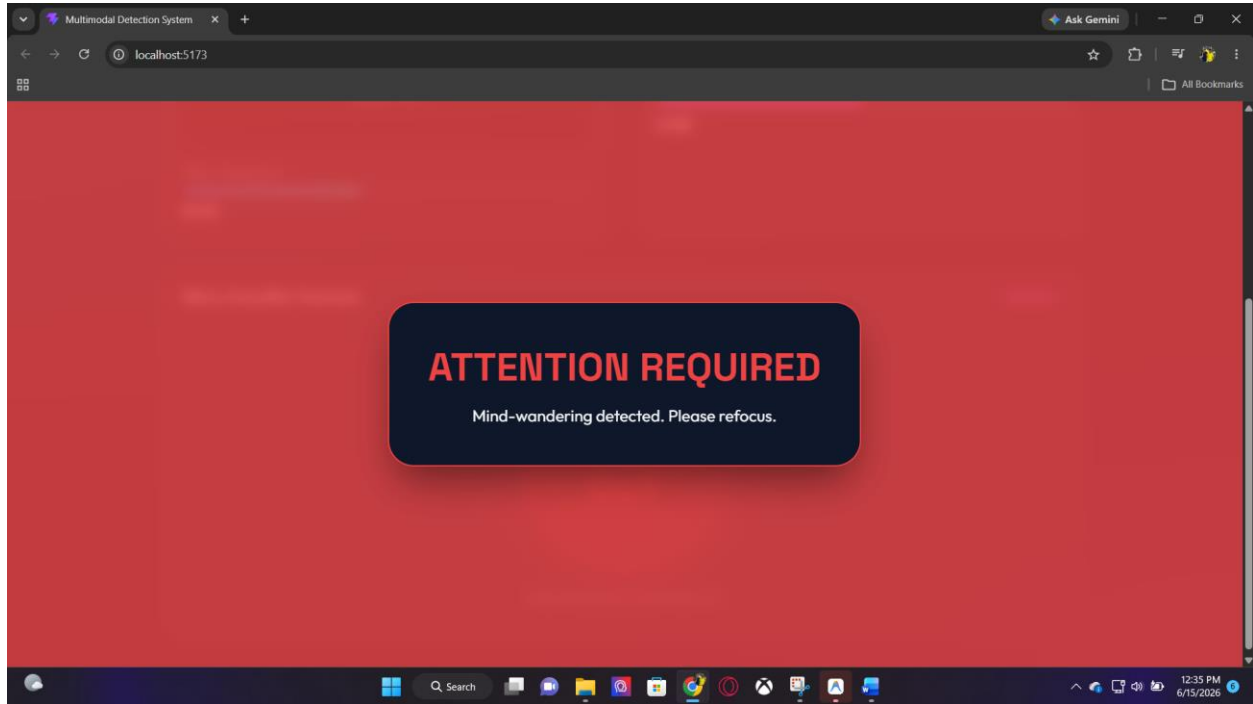
## 5.5 Demo of the Prototype

The system was demonstrated as a functional prototype, successfully classifying Focused, Zoned Out, and User Away states in real time.



## Cognitive Focus Monitor





## 5.6 Chapter Summary

This chapter shows the projects status at the midpoint, confirming that the multimodal approach for zoning out is viable and outlining what is required for final submission.

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